DESCRIPTIVE ANALYTICS REPORT

Netflix Content: Trends, Genres, Ratings & Regional Distribution

Student Name: MUSKAN SRIVASTAVA

Roll Number: 1250258279

Course: BCA DS & AI

Class: BCADS24

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DESCRIPTIVE ANALYTICS REPORT

Netflix Content: Trends, Genres, Ratings & Regional Distribution

Note on data: Netflix live transaction and viewing data is not publicly accessible. Therefore, this project uses a synthetic dataset built to mirror real-world streaming and content-distribution patterns.

This project is based on a structured Netflix-style synthetic dataset created for academic analysis. Category includes Movies and TV Shows. Region covers 12 countries: United States, India, United Kingdom, Japan, South Korea, Spain, France, Canada, Germany, Mexico, Australia and Brazil. Year covers release years 2000–2025, while the content-added records span 2018–2026. Unit of analysis includes count of titles, movie duration in minutes, TV-show duration in seasons, user rating out of 5, and estimated views in millions.

The synthetic dataset is designed to reflect realistic content distribution, audience engagement, and viewing behavior patterns. It is used only for academic descriptive analysis and does not represent Netflix's confidential internal data.

OBJECTIVES

The project applies descriptive analytics to a Netflix-style content dataset. The main objectives are:

1.Compare Movies and TV Shows and understand the content mix.

2. Examine content added over time and identify periods with higher catalog additions.

3. Analyze release-year distribution to understand the age and spread of titles.

4. Identify the most represented genres and countries.

5. Study rating distribution and audience classification patterns.

6. Analyze movie duration and identify the typical runtime range.

7. Examine the relationship between user rating and estimated views.

8. Use cross-tabulation and grouped summaries to find deeper descriptive patterns.

9. Summarize the key metrics and define a practical path from descriptive to predictive analytics.

Key Analytical Questions

What is the overall composition of the catalog?

Which year's show higher or lower content additions?

Which genres and countries are most represented?

Which ratings occur most frequently?

What does the distribution of ratings and movie duration look like?

Do higher ratings and higher estimated views tend to appear together in the sample?

Note: These questions are descriptive: they summarize the dataset and do not claim to predict future demand or prove cause-and-effect relationship

HOW THIS WOULD WORK ON REAL SCALE

Processing a large, constantly changing Netflix content catalog linked to daily viewing and engagement signals is an engineering problem before it is an analytics problem. A realistic architecture for Netflix would ingest content metadata, rating updates, viewing events and regional activity, aggregate them by content type, genre, country and time window, store raw and processed data, and expose the resulting KPIs through dashboards and scheduled reports.

LAYER	PURPOSE	TYPICAL TOOLS
INVESTIGATION	Research, understand and prepare the Netflix content + viewing dataset	ChatGPT/Claude
INGECTION AND STREAM	Ingest content updates, ratings and viewing events; aggregate by content type, genre, region and time	Python pipelines
STORAGE	Retain raw and processed content data, viewing records and daily aggregates	Cloud Data ,MS EXCEL
BATCH AGGREGATION	Daily/periodic KPI computation, genre summaries, regional summaries and anomaly flags	Jupyter
RATING VS VIZ	Present analysis, charts, insights and alerts through dashboards and reports Looker / Tableau / Power BI / MS Word or PDF reports	MS Word or PDF reports

Content Type Analysis

The dataset contains 798 movies and 402 TV shows. Movies account for 66.5% of all records, while TV shows account for 33.5%.

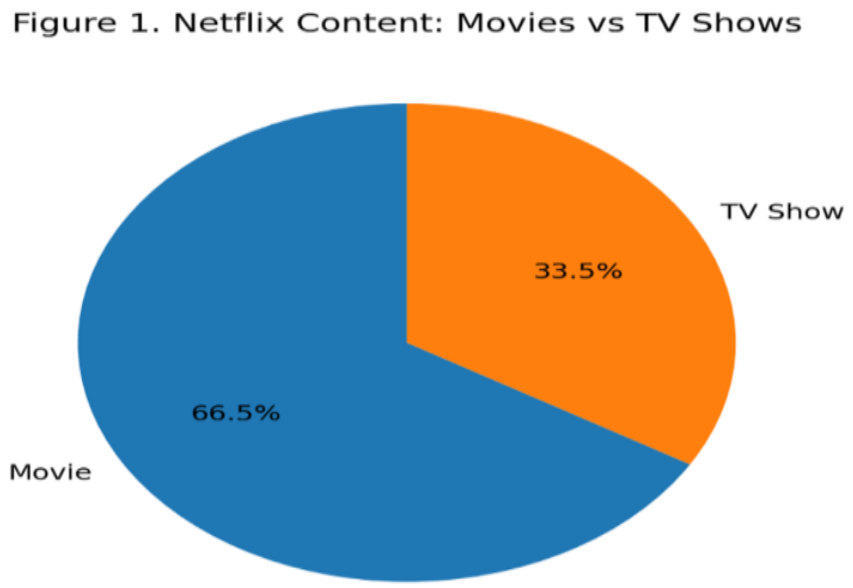


Figure 1. The pie chart shows the part-to-whole composition of the 1,200 records. Movies form the larger share of the synthetic catalog.

Graph Description

The bar chart compares the number of titles by content type. The movie bar is substantially taller than the TV Show bar, showing a larger share of movies.

Insights

Movies account for 798 titles (66.5%), while TV Shows account for 402 titles (33.5%).

The catalog is therefore more heavily represented by movie content in this synthetic dataset.

The difference is useful for planning separate analyses because movie duration is measured in minutes while TV-show duration is measured in seasons.

KPI

1,200 — Total Titles

798 — Movies

402 — TV Shows

66.5% — Movie Share

CONTENT ADDED OVER TIME

The date_added field was converted into a year variable to show how many titles entered the catalog in each year.

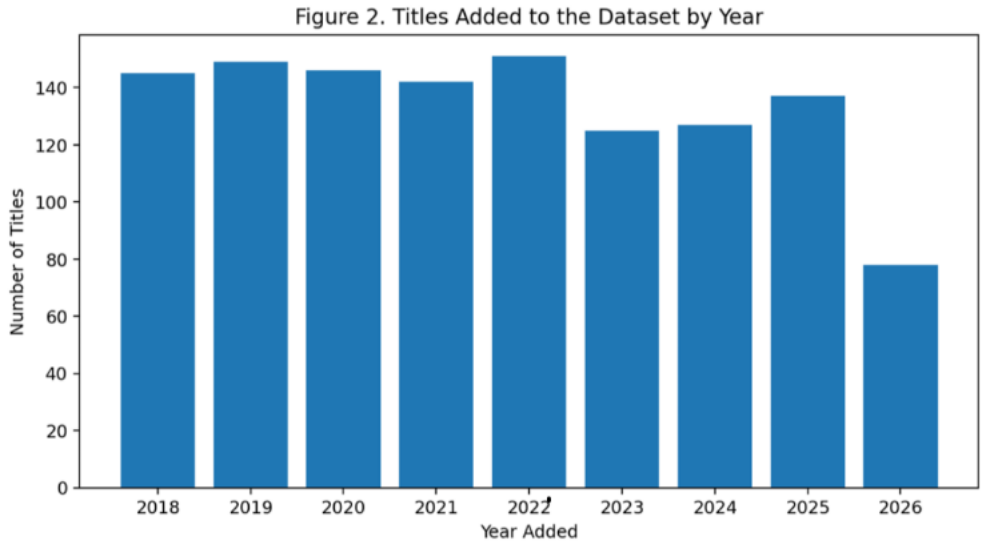


Figure 2. Number of titles added to the synthetic dataset from 2018 to 2026.

Graph Description

The line chart tracks the number of titles added each year. Peaks and dips indicate years in which the catalog receives comparatively more or fewer additions.

Insights

The highest number of content additions occurs in 2022, with 151 titles.

The lowest number occurs in 2026, with 78 titles.

The annual pattern is fairly distributed rather than showing a single extreme spike, suggesting a relatively steady catalog-addition process in the synthetic data.

The year 2026 has fewer additions because the dataset contains only part-year-style records for that latest period.

KPI

1,200 — Added Records

2022 — Peak Year

151 — Peak Additions

78 — Lowest Additions

Release Year Analysis

Graph: Line chart showing number of titles by release year.

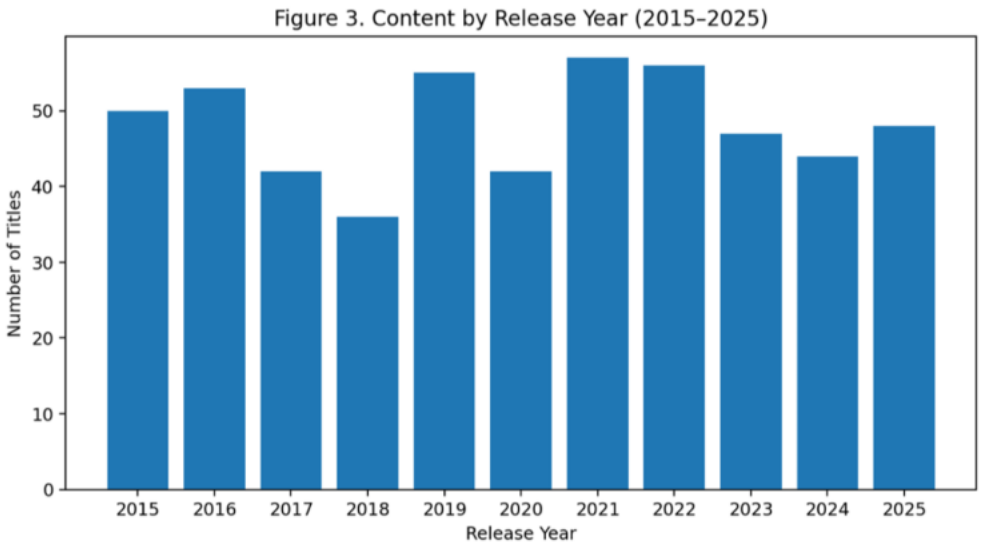


Figure 3. Distribution of titles by original release year for the 2015–2025 period.

Graph Description

The line chart shows the number of titles associated with each release year. It highlights the distribution and identifies years with comparatively high representation.

Insights

The largest number of titles in a single release year is 57 titles in 2021.

The distribution is spread across 2000–2025, indicating a mix of older and newer content.

Several years have similarly high counts, so the catalog is not concentrated in only one release year.

Release year describes when content originated; it is different from the date-added field, which describes when content entered the catalog.

KPI

2000–2025 — Release Range

2021 — Peak Release Year

57 — Peak Titles

1,200 — Total Titles

Genre Analysis

Genre frequency shows which content categories occupy the largest share of the catalog. Drama is the most represented genre with 225 titles, followed by Comedy and Thriller.

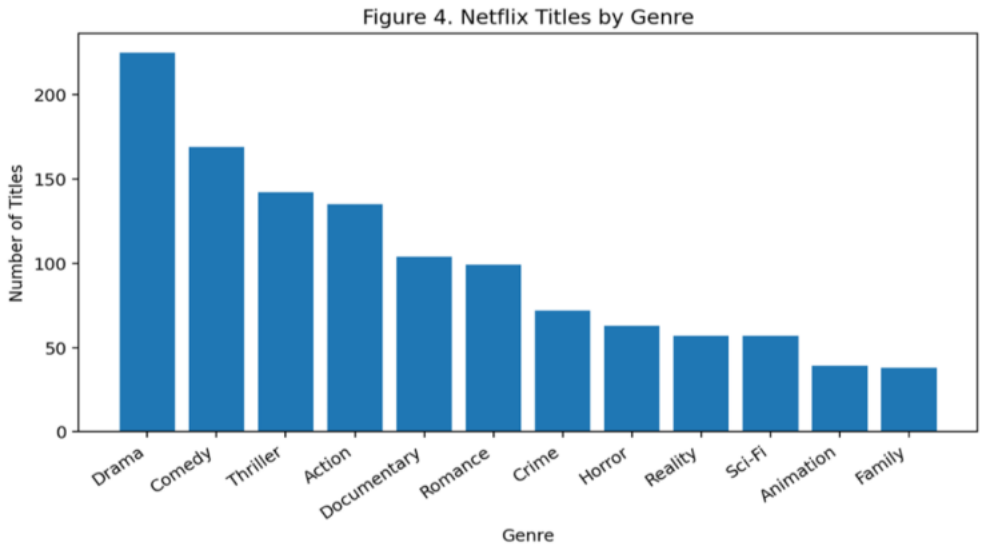


Figure 4. Frequency distribution of the 12 genres represented in the dataset.

Graph Description

The bar chart ranks the ten most common genres by title count. Higher bars represent genres with more titles in the dataset.

Insights

Drama is the most represented genre with 225 titles.

The top five genres are Drama, Comedy, Thriller, Action and Documentary.

Drama and Comedy together form a large portion of the catalog, while Animation and Family are smaller categories in the dataset.

Genre concentration can help identify which content categories dominate the catalog and which categories are comparatively underrepresented.

KPI

Drama — Top Genre

225 — Top Genre Titles

1,200 — Genre Records

11 — Genres

Country-wise Analysis

Country analysis shows the geographic distribution of content. The United States contributes the largest number of titles with 292, followed by India with 219.

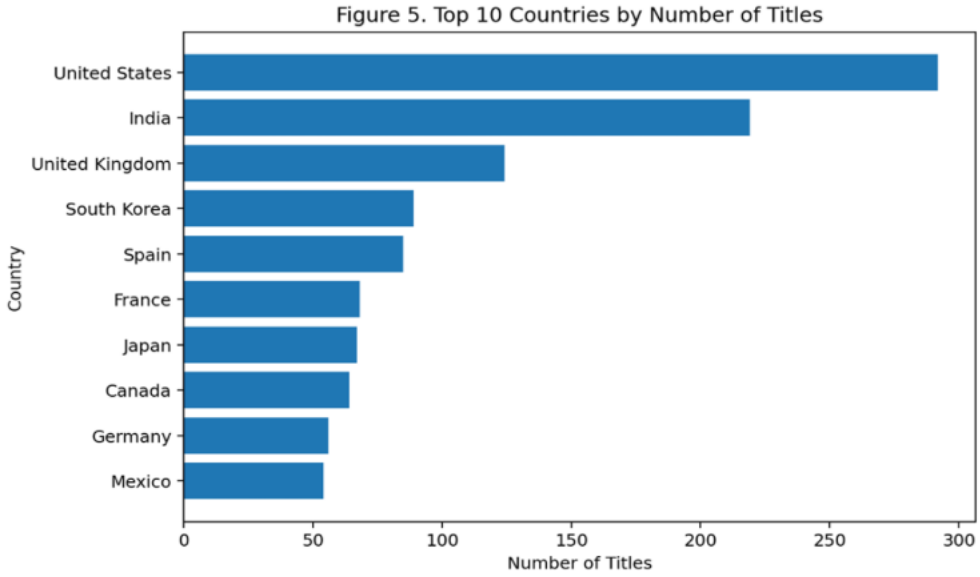


Figure 5. Top 10 countries ranked by number of titles in the dataset.

Graph Description

The bar chart compares the number of titles associated with each of the 12 countries in the synthetic dataset.

Insights

United States has the largest representation with 292 titles.

635 titles come from the top three regions: United States, India and United Kingdom.

The dataset covers 12 regions, demonstrating geographic diversity in content.

Country-wise analysis can support regional catalog planning and comparison of content presence across markets.

KPI

United States — Top Region

292 — Top Region Titles

12 — Regions

635 — Top 3 Titles

Rating Analysis

The most frequent content rating is TV-MA with 325 records, followed closely by TV-14 with 314 records.

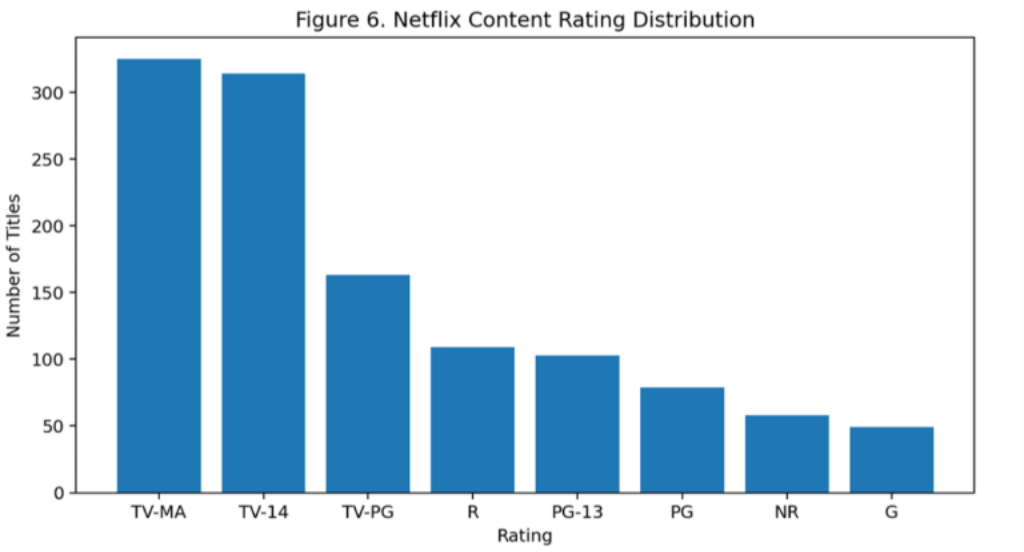


Figure 6. Frequency distribution of content ratings.

Graph Description

The bar chart displays the number of titles under each content-rating code. It shows which audience classifications are most common.

Insights

TV-MA is the most frequent rating with 325 titles.

314 titles are rated TV-14, making it another major audience category.

TV-MA and TV-14 together account for a large share of the dataset, indicating strong representation of mature and teen-oriented content.

Rating codes describe audience suitability; they should not be confused with the numerical user-rating score out of 5

KPI

TV-MA — Most Common Rating

325 — Most Common Count

1,200 — Rated Titles

8 — Rating Codes

Movie Duration Analysis

For the 798 movie records, the average duration is 127.6 minutes, with a median of 128 minutes.

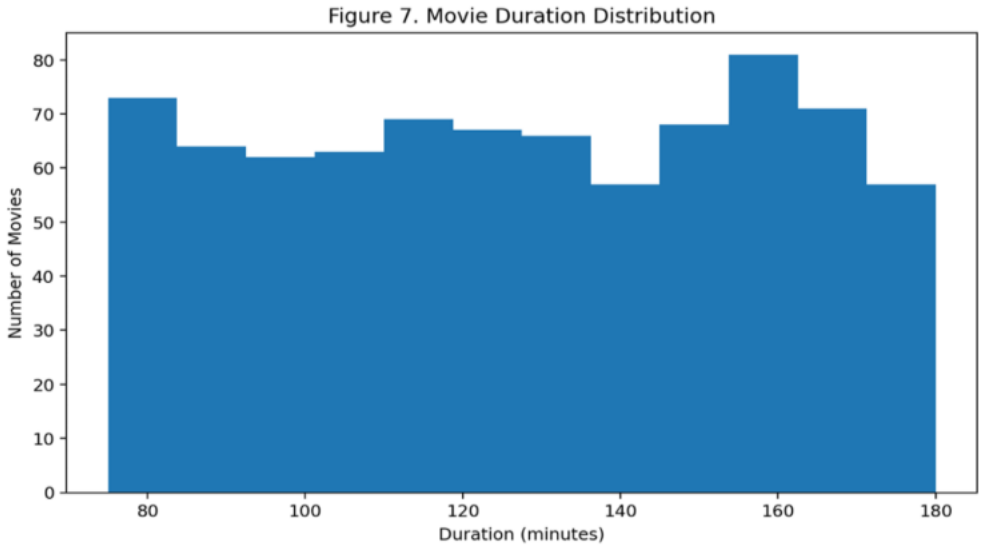


Figure 7. Histogram showing the distribution of movie durations in minutes.

Graph Description

The histogram groups movies by runtime. Taller bars show the duration ranges containing more movies.

Insights

The average movie duration is approximately 127.6 minutes (about 4 hours).

The median duration is 128 minutes (about 4 hours), meaning half of the movies are at or below this value.

Movie durations range from 75 to 180 minutes (about 6 hours).

The distribution is centered around the 2-hour range but includes both shorter and longer movies.

KPI

127.6 min — Average Duration

128 min — Median Duration

75–180 min — Range

30.5 min — Standard Deviation

Rating vs Estimated Views

The scatter plot examines whether user ratings and estimated views tend to move together within the synthetic sample.

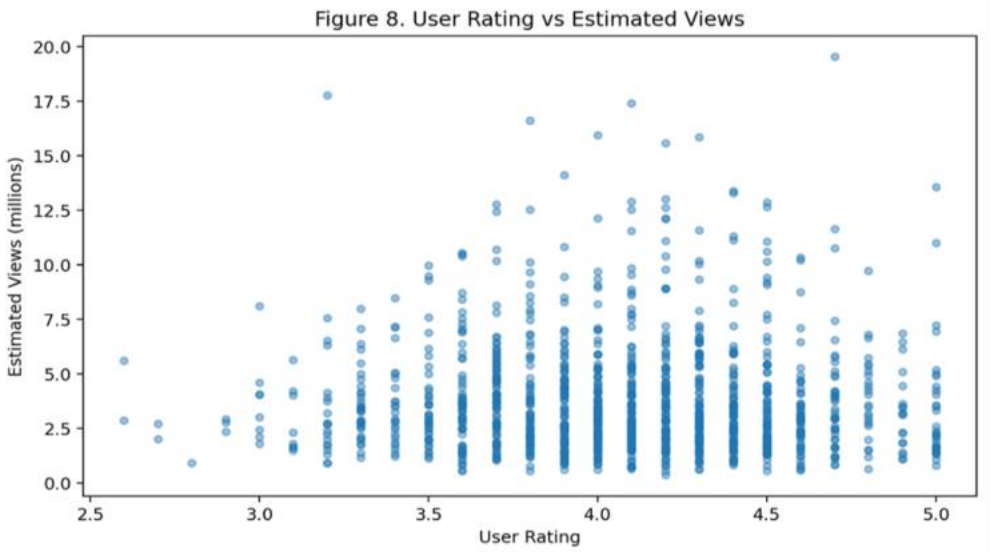


Figure 8. Relationship between synthetic user ratings and synthetic estimated views.

Graph Description

The scatter plot places user ratings on the x-axis and estimated views on the y-axis. The spread of points shows how viewing volume varies across ratings.

Insights

The average user rating is 4.08 out of 5, while the average estimated views are 3.79 million.

The correlation between rating and estimated views is -0.01, indicating a weak relationship in this synthetic dataset.

A rating alone does not explain viewing volume; popularity can also depend on genre, country, promotion, release timing and audience size.

Correlation is descriptive here and does not prove that one variable causes the other.

KPI

4.08/5 — Average User Rating

3.79M — Average Views

19.54M — Maximum Views

-0.01 — Correlation

Cross-Tabulation: Content Type vs Rating

Cross-tabulation compares two categorical variables at the same time. Here, the content type is compared with the content ratings.

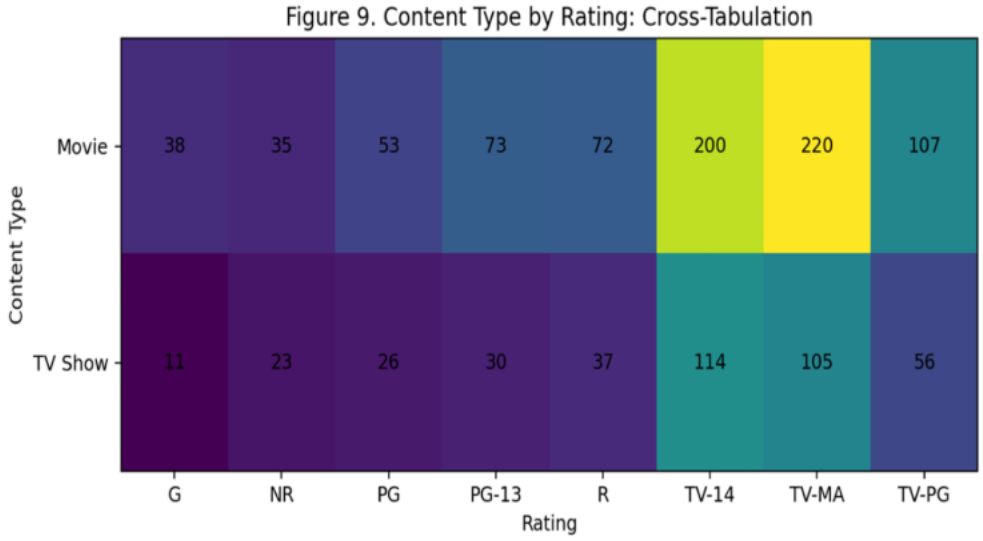


Figure 9. Cross-tabulation of Movies and TV Shows across the eight rating categories.

Graph Description

The grouped bar chart compares Movie and TV Show counts across each rating category, making differences between content types visible.

Insights

Movies have 220 titles rated TV-MA, the highest movie-rating combination in the cross-tabulation.

TV Shows have 114 titles rated TV-14, making it the largest TV Show-rating combination.

The cross-tabulation shows that rating patterns differ by content type, so type and rating should be analyzed together rather than independently.

This two-dimensional view gives a more detailed picture of the catalog than a single rating-frequency chart.

KPI

220 — Movies: TV-MA

114 — TV Shows: TV-14

798 — Movie Total

Average User Rating by Genre

Genre frequency tells us how many titles belong to each category, but it does not show whether those titles have relatively high or low ratings. Average rating by genre provides a second descriptive perspective.

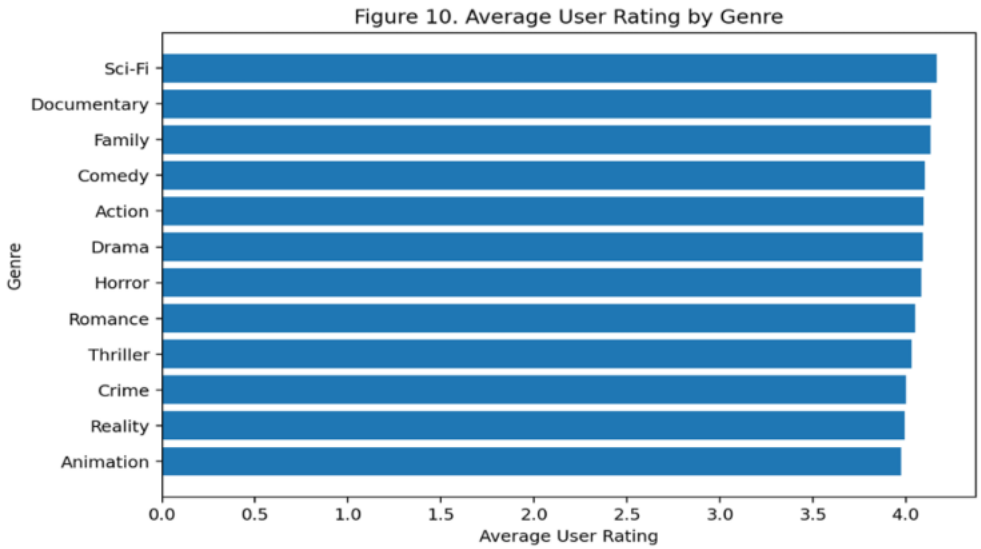


Figure 10. Average synthetic user rating for each genre.

Graph Description

The bar chart ranks genres by their average numerical user rating. Higher bars indicate stronger average audience scores.

Insights

Sci-Fi has the highest average user rating at 4.17/5.

Animation has the lowest average among the listed genres at 3.98/5.

The difference between the highest and lowest genre averages is relatively small, so overall ratings are fairly consistent across genres.

Average genre ratings help compare audience reception at category level rather than title by title.

KPI

Sci-Fi — Highest Average Genre

4.17/5 — Highest Average Rating

Animation — Lowest Average Genre

3.98/5 — Lowest Average Rating

Estimated Views by Country

Country-level estimated views provide a second geographic measure beyond the number of titles. This helps distinguish countries with many titles from countries whose records collectively have higher estimated views.

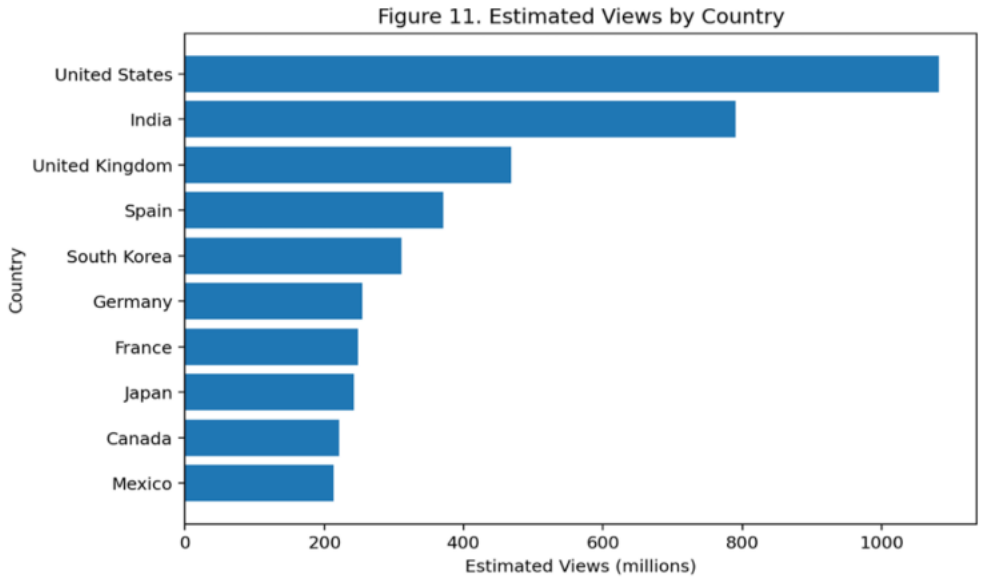


Figure 11. Total synthetic estimated views by the top 10 countries.

Graph Description

The bar chart compares the total estimated views generated by titles associated with each country.

Insights

United States has the highest estimated views at 1082.65 million.

Brazil has the lowest estimated views at 132.32 million.

Higher views generally align with countries that also have larger numbers of titles, but the relationship is not identical.

Country-level views can support regional performance comparison and can be combined with title counts to derive views-per-title metrics.

KPI

United States — Highest Views

1082.65M — Top Country Views

132.32M — Lowest Country Views

Key Matrics

METRICS	VALUE
TOTAL TITLE	12000
MOVIES	798
TV SHOWS	402
REGIONS	12
GENRE	11
RELEASE YEAR RANGE	2000-2005
AVERAGE USER RATING	4.08/5
TOTAL ESTIMATED VIEWS	5,545.79 MILLION
AVERAGE MOVIE DURATION	127.6 MINUTES
MOST COMMON RATING	TV-MA –325 TITLES
TOP GENRE	DRAMA –225 TITLES
TOP REGION BY TITLE	UNITED STATE- 292 TITLES
TOP REGION BY VIEW	UNITED STATES- 1082.65

NEXT STEP: MOVE FROM DESCRIPTIVE TO PREDICTIVE

Descriptive analytics explains what is present in the dataset. The next stage is to use the same variables to predict future outcomes. For this Netflix-style dataset, predictive analytics can be designed around estimated views, popularity, rating or content demand.

1. Define the Prediction Target: A practical target is estimated views in millions. Another option is to classify titles into high-view and low-view groups.

2. Candidate Input Variables:

Content type, genre, country, release year, date added, duration, rating and user rating can be transformed into model-ready features.

3. Feature Engineering

Create variables such as: Content age, Movie length bands, Number of seasons, Genre indicators, Countryry indicators,Historical average ratings

4. Possible Models

Linear Regression can predict a numeric view value.

Decision Trees and Random Forests can predict high/low popularity.

More advanced models can be evaluated after sufficient real data is available.

5. Evaluation

For regression:

MAE

RMSE

R^2

For classification:

Accuracy

Precision

Recall

F1-score

6. Business Use

A predictive system could support:

Content acquisition

Regional demand planning

Promotion priorities

Personalized recommendations

Important Limitation

Because this project uses synthetic data, predictive results would be illustrative. A production model would require reliable historical viewing and engagement data.